

Bayesian Functional Analysis of Spatial Temperature Profiles

Group 11

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Abstract

This report studies Bayesian functional models for modelling spatial temperature data. We first introduce a one-dimensional framework to present the methodology using B-splines, and then extend the approach to a two-dimensional spatial setting using a linear finite element method basis. In both cases, we use a Bayesian random-effects model, where the curves are represented as linear combinations of random coefficients and basis functions—B-splines in the one-dimensional case and linear finite element basis functions in the two-dimensional case. We use Monte Carlo Markov Chain methods to compute the posterior means, more specifically a Gibbs sampler and Random Walk Metropolis-Hastings.

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1 Introduction and Project Background

Temperature measurements are typically recorded as discrete observations at specific locations and time points. In this project, we are provided with a global dataset of Earth’s surface temperatures consisting of monthly mean observations in degrees Celsius over the period from 1983 to 2007. While the data are observed discretely, it is often desirable to approximate an underlying continuous function over time and space in order to enable deeper analysis, interpretation, and prediction.

Rather than relying on classical numerical techniques, such as least squares estimation or other standard deterministic approximation methods, we adopt a statistical modeling perspective. Similar to traditional basis function approaches, we represent the temperature field as a linear combination of basis functions and seek to determine the optimal coefficients. However, instead of estimating these coefficients purely through deterministic optimization, we embed the problem within a hierarchical Bayesian framework.

In this framework, year-specific temperature profiles are modeled as random deviations from a common population mean. This structure allows the model to “borrow strength” across the 25 years of observations (Thompson and Rosen 2008), effectively separating the stable global spatial trend from year-specific variability while smoothing transient noise. The hierarchical formulation provides a principled way to quantify uncertainty and incorporate variability at multiple levels.

We apply this methodology to model the Earth’s surface temperature profile over selected longitude and latitude ranges. As a first step, we construct functional models for a one-dimensional spatial transect at a fixed latitude using B-spline basis functions. After validating this approach, we extend the framework to a two-dimensional spatial domain using Finite Element Method (FEM) basis functions. This mixed-effects formulation enables robust estimation of the spatial temperature process while accounting for inherent variability and measurement noise in the dataset. Finally, we investigate the computational complexity associated with the FEM-based approach, particularly when incorporating a Gaussian neighborhood structure to model spatial dependence.

2 Data Description

We consider a global surface temperature dataset consisting of monthly mean temperatures (measured in degrees Celsius) over the period 1983–2007. In the scope of this project, we aggregate the monthly observations and work with yearly mean temperatures only. Let $i \in \{0, \dots, 24\}$ index the year, where $i = 0$ corresponds to 1983 and $i = 24$ to 2007. The spatial domain is described by longitude–latitude coordinates. We denote longitude by x and latitude by y as in (1) :

$$x \in [-180^\circ, 180^\circ], \quad y \in [-90^\circ, 90^\circ]. \quad (1)$$

Mathematically, the temperature field in year i can be interpreted as a discrete function:

Definition 2.1 (Yearly Temperature Field). *Let $\mathcal{D}$ denote the set of observed longitude-latitude grid points, i.e.,*

$$\mathcal{D} = \{(x_j, y_p) \mid x_j \in [-180^\circ, 180^\circ], y_p \in [-90^\circ, 90^\circ]\}. \quad (2)$$

The yearly temperature field in year i is defined as the discrete function

$$T_i : \mathcal{D} \rightarrow \mathbb{R}, \quad (x_j, y_p) \mapsto T_i(x_j, y_p), \quad (3)$$

where $T_i(x_j, y_p)$ denotes the yearly mean temperature at spatial location (x_j, y_p) in year i .

In practice however, since the entire span of the globe is too broad, we use only a finite number of observations as in spatial grid points. Let (4) denote the observed longitude and latitude coordinates.

$$\{x_1, \dots, x_J\} \subset [-180^\circ, 180^\circ], \quad \{y_1, \dots, y_P\} \subset [-90^\circ, 90^\circ] \quad (4)$$

Thus, for each year i and each grid point (x_j, y_p) , we observe $T_i(x_j, y_p) \in \mathbb{R}$. This discrete representation serves as the basis for constructing a continuous functional approximation of the spatial temperature field.

2.1 1D case

For convenience and to build familiarity with the dataset, we begin by considering a one-dimensional spatial subset. Specifically, we fix a latitude and examine the temperature variation along a transect of longitudes. In this work, we fix the latitude at the equator, $y_p = 0$, and focus on the longitudinal range from 30°E to 60°E . Let (5) denote the subset of longitude grid points in this range, where 30 is the number of discrete longitude points considered.

$$D_{x_1} = \{x_j \in [30^\circ, 60^\circ], \quad j = 1, \dots, 30\} \quad (5)$$

For each year i , the temperature profile along this longitudinal transect can be interpreted as a discrete function as in (6), according to Definition 2.1

$$T_i(\cdot, 0) : D_{x_1} \times 0 \rightarrow \mathbb{R}, \quad x_j \times 0 \mapsto T_i(x_j, 0), \quad j = 1, \dots, 30, \quad (6)$$

where $T_i(x_j, 0)$ represents the yearly mean temperature at longitude x_j and latitude $y_p = 0$ in year i . For simplicity, we may also write $T_i(x_j) := T_i(x_j, 0)$, $j = 1, \dots, 30$, to emphasize the one-dimensional longitudinal transect. In this sense, each year corresponds to a distinct discrete curve along the longitude domain, which we will approximate by a continuous functional representation in the subsequent modelling.

For illustrative purposes, the temperature data along the specified longitudinal range at the equator is visualized in Figure 1. Here, "Year" is an index with $i = 1$ corresponding to 1983, increasing annually to $i = 25$, corresponding to 2007.

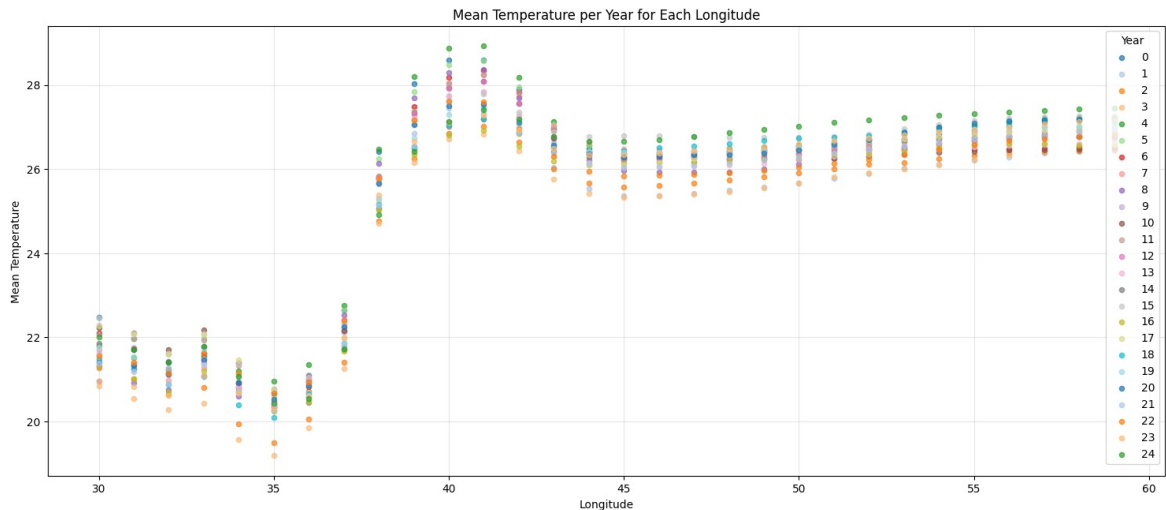


Figure 1: Yearly mean temperature for period 1983–2007 for the subgrid D_{x_1}

2.2 2D case

After considering the one-dimensional case, we extend our analysis to a two-dimensional spatial subset. Specifically, we focus on the geographical region bounded by longitudes 2° – 22° and latitudes 33° – 53° . Let the subsets of longitude and latitude grid points in this range in be denoted as in (7), where 20×20 are the number of discrete longitude and latitude points, respectively.

$$D_x = \{x_j \in [2^\circ, 22^\circ], \quad j = 1, \dots, 20\}, \quad D_y = \{y_p \in [33^\circ, 53^\circ], \quad p = 1, \dots, 20\} \quad (7)$$

For each year i , the temperature field of this subgrid can be regarded as a discrete function as in (8) according to Definition 2.1, where $T_i(x_j, y_p)$ represents the yearly mean temperature at longitude x_j and latitude y_p in year i .

$$T_i : D_x \times D_y \rightarrow \mathbb{R}, \quad (x_j, y_p) \mapsto T_i(x_j, y_p), \quad j = 1, \dots, 20, \quad p = 1, \dots, 20, \quad (8)$$

In practice, this discrete function serves as the basis for constructing a continuous functional approximation of the spatial temperature field over the selected two-dimensional domain.

For illustrative purposes, the temperature data over this two-dimensional geographical region is visualized in Figure 2.

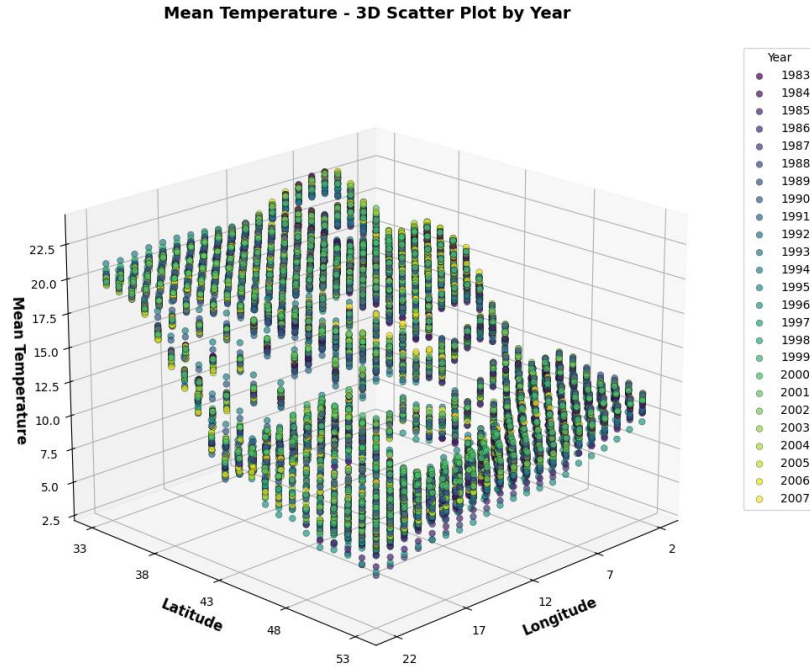


Figure 2: Yearly mean temperature for period 1983 - 2007 for the subgrid $D_x \times D_y$

3 One-Dimensional Model

3.1 Spline Basis

Our goal is to estimate the underlying smooth spatial temperature profile for each year and understand the common spatial patterns across all years. Therefore, we want to find sufficiently smooth functions f_i such that the model can be stated as in (9)

$$f_i(x_j) = T_i(x_j) + e_{ij}, \quad i = 1 \dots T, \quad j = 1 \dots J \quad (9)$$

with the errors e_{ij} being as small as possible, i being the considered year and x_j the data point at location j .

To do this we use *B-splines*, which are piecewise polynomial basis functions, where each individual basis function has its own domain, providing local control and flexibility. We use cubic B-splines, which produce smooth curves $f_i(x)$ that are twice differentiable. The longitude interval $[30, 60]$ is split into 5 equidistant subintervals, resulting in $K = 8$ B-spline basis functions. The functional model looks as in (10)

$$f_i(x) = \sum_{k=1}^8 \beta_k B_k(x) + \sum_{k=1}^8 b_{ik} B_k(x) + \varepsilon_i(x) \quad (10)$$

$$\text{Matrix notation: } \mathbf{f}_i = X_i \boldsymbol{\beta} + X_i \mathbf{b}_i + \boldsymbol{\varepsilon}_i$$

Here, $B_k(x)$ are the B-spline basis functions, X_i is the corresponding design matrix for year i , where $[X_i]_{jk} = B_k(t_{ij})$, $\boldsymbol{\beta}$ is the mean functional outcome, and $\mathbf{b}_i$ is the “large-scale” deviation of the i -th year’s functional outcome from the mean, i.e. random effect.

3.2 Bayesian Model

In the hierarchical Bayesian framework, the followings prior specifications are considered.

- **Fixed Effects ($\boldsymbol{\beta}$):** The mean spatial profile.

$$\boldsymbol{\beta} \sim \text{MVN}(\mathbf{0}, c\mathbf{I}_K)$$

- **Random Effects ($\mathbf{b}_i$):** The year-specific deviation from the mean.

$$\mathbf{b}_i \stackrel{iid}{\sim} \text{MVN}(\boldsymbol{\beta}, \boldsymbol{\Sigma}_b), \quad \text{for } i = 0, \dots, 24$$

- **Covariance Matrix ($\boldsymbol{\Sigma}_b$):**

$$\boldsymbol{\Sigma}_b \sim \text{IW}(\eta_b, \eta_b \mathbf{S}_b)$$

- **Overall Model Error Term ($\boldsymbol{\varepsilon}_i$):**

$$\boldsymbol{\varepsilon}_i \stackrel{iid}{\sim} N(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}), \quad \text{for } i = 0, \dots, 24$$

- **Residual Noise (σ_ε^2):** Measurement error variance.

$$\sigma_\varepsilon^2 \sim \text{IG}(c_\varepsilon, d_\varepsilon)$$

Where all hyper parameters are kept vague to ensure that the data dominates the posterior distributions.

Since the joint posterior distribution is analytically intractable, we use a Gibbs Sampler to estimate the parameters. The algorithm iteratively samples from the full conditional distributions of each parameter, updating them sequentially, based on the current values of the other parameters.

At each iteration of the MCMC sampler:

1. **Update the mean spatial profile $\boldsymbol{\beta}$:**

$$\boldsymbol{\beta} \mid \mathbf{b}, \sigma_\varepsilon^2, y \sim N(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta),$$

with

$$\boldsymbol{\Sigma}_\beta = \left(\sum_i \frac{X_i^T X_i}{\sigma_\varepsilon^2} + \frac{\mathbf{I}}{c} \right)^{-1}, \quad \boldsymbol{\mu}_\beta = \boldsymbol{\Sigma}_\beta \sum_i \frac{X_i^T (y_i - X_i \mathbf{b}_i)}{\sigma_\varepsilon^2}.$$

2. **Update the random effects** $\mathbf{b}_i$, for all i :

$$\mathbf{b}_i \mid \boldsymbol{\beta}, \sigma_\varepsilon^2, \Sigma_b, \mathbf{y}_i \sim N(\boldsymbol{\mu}_{b_i}, \Sigma_{b_i}),$$

where

$$\Sigma_{b_i} = \left(\frac{X_i^T X_i}{\sigma_\varepsilon^2} + \Sigma_b^{-1} \right)^{-1}, \quad \boldsymbol{\mu}_{b_i} = \Sigma_{b_i} \frac{X_i^T (\mathbf{y}_i - X_i \boldsymbol{\beta})}{\sigma_\varepsilon^2}.$$

3. **Update the residual variance** σ_ε^2 :

$$\sigma_\varepsilon^2 \mid \mathbf{y}, \boldsymbol{\beta}, \mathbf{b} \sim IG \left(c_\varepsilon + \frac{1}{2} \sum_i m_i, d_\varepsilon + \frac{1}{2} \sum_i \|\mathbf{y}_i - X_i(\boldsymbol{\beta} + \mathbf{b}_i)\|^2 \right).$$

4. **Update the random-effects covariance** Σ_b :

$$\Sigma_b \mid \mathbf{b} \sim IW \left(\eta_b + n, \eta_b S + \sum_i \mathbf{b}_i \mathbf{b}_i^T \right).$$

3.3 Posterior Distribution

The posterior means across R retained samples are calculated as :

$$\hat{\mu}(t) = \frac{1}{R} \sum_{r=1}^R \mathbf{B}(t)^T \boldsymbol{\beta}^{(r)}, \quad (11)$$

$$\hat{f}_i(t) = \frac{1}{R} \sum_{r=1}^R \mathbf{B}(t)^T \mathbf{b}_i^{(r)}. \quad (12)$$

Pointwise credible intervals for $\mu(t)$ or $f_i(t)$ are obtained from empirical quantiles of these sampled values.

3.4 Computational Approach and Implementation

The code for the model can be found in this GitHub repository. The overall structure of the code is as follows:

- **BSpline.py** – A wrapper for generating and evaluating B-Spline basis. It is used for generating the splines for the one-dimensional case.
- **conditionals.py** – contains the conditional distributions for $\mathbf{b}$, $\boldsymbol{\beta}$, σ_ε^2 , and Σ_b . It also includes a Metropolis-Hastings algorithm to sample σ_b in the two-dimensional case, where a Gaussian neighbourhood structure with a Matérn covariance matrix is used instead of Inverse Wishart.
- **data_import.py** – Handles the import of the mean temperature data for latitude and longitudes. It includes a function for one dimensional data, where the points are chosen from the equator, ranging from longitudes 30° to 60° . Another function is implemented to import 2D data with longitude values from 2° to 22° and latitude values from 33° to 53° .
- **FEMBasis.py** Generates and evaluates a finite element basis for the spatial data, utilizing triangulation. It is used for the 2D scenario to form a basis for the varying temperature data over the latitude and longitude values.
- **main.py** Forms the BSpline basis and priors to run Monte Carlo Markov Chain simulations. The posterior values are then used to create a predictive model for the mean temperatures over the locations.

- **MCMC.py** Uses a Gibbs sampler to draw samples for b , β , σ_e^2 , and Σ_b from their posterior distributions for a certain amount of iterations (default value is 2000) and stores the samples after burn-in (default value is 1000) to be used for posterior inference.
- **MCMC_MH.py** Follows the same logic as **MCMC.py**, however, uses instead the Metropolis-Hastings algorithm to sample σ_b . This MCMC approach is preferred when two-dimensional data are analysed.
- **sigma_distribution.py** Creates a posterior density function for Σ_b , using a Matérn covariance matrix with $\nu = 3/2$ with lognormal priors σ_b^2 and ρ . The distribution is used in order to create a sparse matrix, which works more efficiently than Inverse-Wishart for two-dimensional data.

3.5 Results and Discussion

Figure 3 presents the mean posterior (red line) and the posterior samples for all 25 years (blue lines). The mean curve accurately captures the overall longitudinal trend, while the posterior samples remain closely concentrated around it. The small spread between the red and blue curves reflects reduced uncertainty due to the relatively large number of groups included in the model.

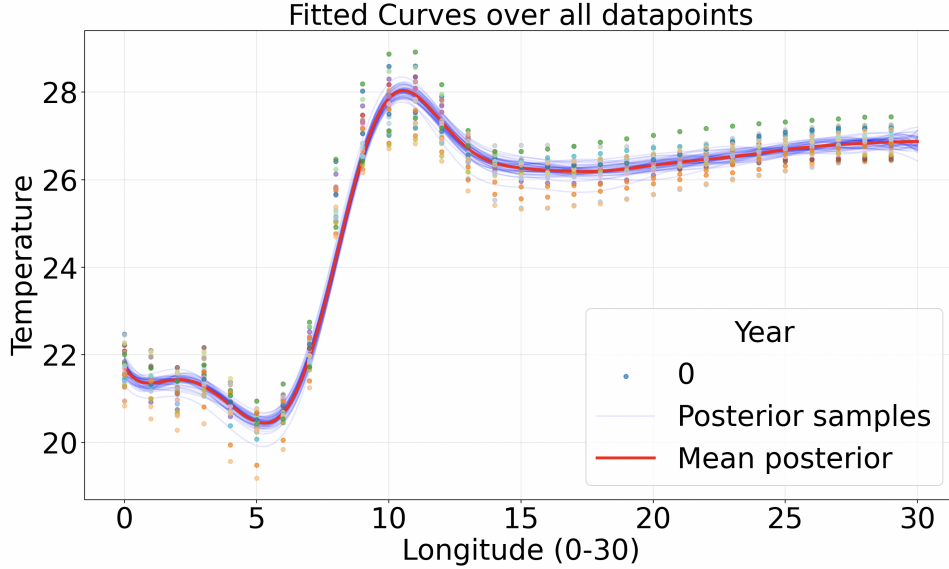


Figure 3: Posterior Mean ($\hat{\mu}(x) = \mathbf{B}(x) \left(\frac{1}{R} \sum_{r=1}^R \mathbf{b}_i^{(r)} \right)$)

Figure 4 presents the fitted curves and trace plots for the model parameters. The top panels show the posterior mean curve for all b_i 's (corresponding to the 25 years), together with the year-specific posterior means for Year 0 (top left) and Year 1 (top right). In both cases, the fitted curves closely follow the observed data, indicating a good model fit at the individual-year level.

The trace plots in the bottom-left panel indicate that, after burn-in, the Markov chains stabilize around constant mean levels, with no visible trends or drifts. This behavior suggests satisfactory convergence and relatively low posterior uncertainty for the parameters.

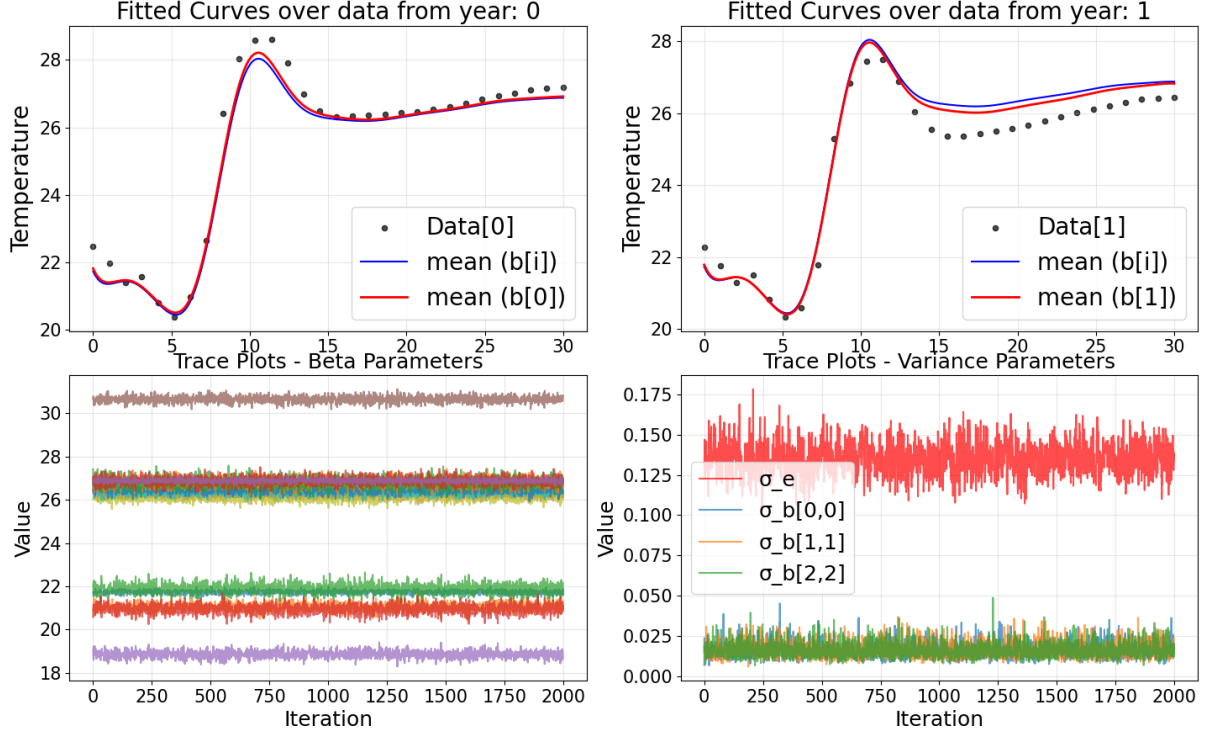


Figure 4: Individual year profiles (top) and trace plots (bottom).

4 Two-Dimensional Model

The primary goal of this project is to extend the one-dimensional functional model to a two-dimensional spatial setting. The one-dimensional framework was developed as a preparatory step and now serves as the foundation for constructing the two-dimensional model.

4.1 Extension to Two Dimensions

We extend the functional model by allowing the underlying process to depend on two spatial coordinates, namely longitude and latitude. Let (x, y) denote a location in the spatial domain. The functional observation for subject i is modeled as

$$f_i(x, y) = \sum_{k=1}^K \beta_k B_k(x, y) + \sum_{k=1}^K b_{ik} B_k(x, y) + \varepsilon_i(x, y), \quad (13)$$

where $\{B_k(x, y)\}_{k=1}^K$ denotes a set of two-dimensional basis functions, β are the fixed effects, b_i are subject-specific random effects, and $\varepsilon_i(x, y)$ is a stochastic error term.

In matrix notation, the model can be written as

$$\mathbf{f}_i = X_i \boldsymbol{\beta} + X_i \mathbf{b}_i + \boldsymbol{\varepsilon}_i, \quad (14)$$

where X_i is the design matrix constructed from evaluations of the basis functions.

Our modeling goal is to approximate the observed spatial field $T_i(x_j, y_k)$ at discrete spatial locations (x_j, y_k) by

$$f_i(x_j, y_k) = T_i(x_j, y_k) + \varepsilon_{ijk}. \quad (15)$$

To achieve this, a suitable two-dimensional basis is required. Unlike the one-dimensional case, where B-splines were used, we now employ a finite element basis, which naturally extends to two-dimensional domains.

4.2 Finite Element Basis

We use a **linear finite element method (FEM) basis**, consisting of piecewise linear functions defined over a triangulation of the spatial domain.

Definition 4.1 (Linear Finite Element Basis). *Let $\{\phi_j(x, y)\}_{j=1}^K$ denote a linear finite element (FEM) basis defined over a triangulation of a two-dimensional spatial domain. Each basis function $\phi_j(x, y)$ is a hat function associated with node j , satisfying*

$$\phi_j(x_k, y_k) = \begin{cases} 1, & \text{if } j = k, \\ 0, & \text{otherwise.} \end{cases}$$

The functions are piecewise linear within each triangle of the mesh and have local support.

In our setting, the spatial domain is defined as a rectangular region over longitude and latitude. A fixed (square) number of nodes is selected within this domain, and a triangulation is constructed to partition the region into triangular elements. The corresponding two-dimensional FEM basis functions are then generated from this triangulation and used to represent the spatially varying functional data.

This finite element representation provides a flexible and computationally efficient basis for modelling spatially varying functional data in two dimensions.

4.3 Initial Exploration and Challenges

The main issue with the implementation of FEM basis functions in the two-dimensional scenario is the computational complexity. Let n_J and n_P be the numbers of grid points along longitude and latitude, respectively. The total number of FEM basis functions is then $K = n_J \times n_P$. Directly adopting the Inverse Wishart prior as in the 1D case causes computational problems: MCMC iterations require repeated inversions of the large dense $K \times K$ Σ_b matrices, with $O(K^3)$ cost. To deal with this, we impose a Gaussian neighbourhood approximation with a Matérn covariance kernel, which yields a sparse matrix, where each node only interacts with its local neighbours. This sparse matrix reduces computational complexity, making Bayesian inference feasible in the 2D case.

4.4 Gaussian Neighbourhood Structure for the Covariance Matrix

When all spatial observations are stacked into a single vector, the associated spatial covariance matrix has dimension equal to the number of grid points squared. For realistic global grids, this dimension can easily reach several thousands or even tens of thousands. Consequently, the covariance matrix becomes extremely large, making its estimation, storage, and inversion computationally demanding.

To overcome this difficulty, one may impose a Gaussian neighborhood (Sangalli, J. O. Ramsay, and T. O. Ramsay 2013) structure on the spatial covariance. Under this assumption, the covariance between two locations depends only on their geographical distance and decreases smoothly as the distance increases. This reflects the natural expectation that nearby locations exhibit stronger dependence, while locations far apart are only weakly correlated. (Datta et al. 2016)

An important computational consequence of this structure is that the Gaussian kernel decays rapidly with distance (Finley, Datta, and Banerjee 2022). As a result, correlations between sufficiently distant locations become negligible. In practice, these very small values can be truncated, yielding a covariance matrix that is approximately sparse. Alternatively, one can work with the corresponding precision matrix (the inverse covariance), which under a neighborhood formulation often has an explicitly sparse structure. Sparse matrices require substantially less storage and allow the use of efficient numerical linear algebra routines. This significantly reduces

the computational burden of posterior inference and sampling while maintaining a realistic and smoothly varying spatial representation of the temperature surface.

4.5 Random Effects Matrix in 2D Case

Unlike in the one-dimensional case, we adopt a Matérn covariance structure for Σ_b . The Matérn class provides an interpretable parameterization of spatial dependence through a small number of hyperparameters. This significantly reduces the dimensionality of the covariance specification while allowing us to model realistic spatial correlation patterns. As a result, the Matérn formulation offers improved scalability and greater flexibility for modeling two-dimensional spatial temperature fields.

We first define Covariance Matrix with the Matérn Kernel, which leads to:

$$\Sigma_{ij}(\sigma_b^2, \rho, \nu) = \sigma_b^2 \cdot M_\nu \left(\frac{d_{ij}}{\rho} \right) \quad (16)$$

Given the distance between 2 points d_{ij} we can compute the covariance. In order to reduce complexity we fix $\nu=3/2$. For the variable ρ and σ_b^2 we choose the following distributions:

$$p(\Sigma_b) = \pi(\sigma_b^2) \cdot \pi(\rho)$$

The priors for the hyperparameters are:

$$\sigma_b^2 \sim \pi(\sigma_b^2) \quad (\log \text{ normal}(1,1)) \quad (17)$$

$$\rho \sim \pi(\rho) \quad (\log\text{-normal}(1,1)) \quad (18)$$

The posterior of Σ_b is not semi-conjugate anymore. In order to compensate this, we use a Random Walk Metropolis-Hastings (RW-MH) inside the Gibbs Sampler to sample from the distribution of Σ_b .

4.6 Metropolis–Hastings Algorithm

First of all we find the posterior probability of our Σ_b to be:

$$\begin{aligned} p(\Sigma_b \mid \mathbf{y}, \boldsymbol{\beta}, \mathbf{b}, \sigma_\varepsilon^2) &\propto p(\mathbf{y} \mid \mathbf{b}, \sigma_\varepsilon^2) \cdot p(\mathbf{b} \mid \boldsymbol{\beta}, \Sigma_b) \cdot p(\boldsymbol{\beta}) \cdot p(\sigma^2) \cdot p(\Sigma_b) \\ &\propto p(\mathbf{b} \mid \boldsymbol{\beta}, \Sigma_b) \cdot p(\Sigma_b) \\ &\propto \underbrace{|\Sigma_b(\rho, \sigma_b^2)|^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{b}_i - \boldsymbol{\beta})^T (\Sigma_b(\rho, \sigma_b^2))^{-1} (\mathbf{b}_i - \boldsymbol{\beta}) \right]}_{p(\mathbf{b} \mid \boldsymbol{\beta}, \Sigma_b)} \\ &\quad \cdot \underbrace{\frac{1}{\sigma_b^2} \exp \left(-\frac{(\ln \sigma_b^2 - 1)^2}{2} \right)}_{p(\sigma^2)} \cdot \underbrace{\frac{1}{\rho} \exp \left(-\frac{(\ln \rho - 1)^2}{2} \right)}_{p(\rho)} \end{aligned}$$

Then random Walk Metropolis–Hastings algorithm for updating (ρ, σ_b^2) consists of the following:

1. **Current state:** $(\rho^{(t)}, \sigma_b^{2(t)})$.
2. **Propose new state:**

$$(\rho^*, \sigma_b^{2*}) = (\rho^{(t)}, \sigma_b^{2(t)}) + (\epsilon_\rho, \epsilon_{\sigma^2}), \quad \epsilon_\rho, \epsilon_{\sigma^2} \sim \mathcal{N}(0, 1)$$

3. **Acceptance ratio:**

$$\alpha = \frac{p(\Sigma_b(\rho^*, \sigma_b^{2*}) \mid \mathbf{y}, \boldsymbol{\beta}, \mathbf{b}, \sigma_\varepsilon^2)}{p(\Sigma_b(\rho^{(t)}, \sigma_b^{2(t)}) \mid \mathbf{y}, \boldsymbol{\beta}, \mathbf{b}, \sigma_\varepsilon^2)}$$

4. Update:

$$(\rho^{(t+1)}, \sigma_b^{2(t+1)}) = \begin{cases} (\rho^*, \sigma_b^{2*}), & \text{with probability } \min(1, \alpha), \\ (\rho^{(t)}, \sigma_b^{2(t)}), & \text{otherwise.} \end{cases}$$

4.7 Results and Interpretation

4.7.1 General results for posterior samples

Figures 5 and 6 include the fitted surfaces and trace plots for the beta and variance parameters for the two-dimensional model. Both the fitted surfaces for the Inverse Wishart and Metropolis-Hastings resemble each other. Similar to what we observed in 1D case, the the posterior means closely follows the average of the observed data, indicating a good model.

In the bottom left plots, we observe the values of the first 10 beta parameters at each iteration after the burn-in point. All of the parameters oscillate around a constant value, indicating that the beta parameters successfully converged for the model. Also, the beta values in Figures 5 and 6 have similar values. However, for variance parameters, the trace plots look different from each other. In Figure 5, a similar convergent structure is seen for the values Σ_b and σ_e . However, in Figure 6, the Σ_b value has not converged to a constant value. That is because the Metropolis-Hastings random walk algorithm was used to sample posterior values for Σ_b . The values move between 0.25 and 0.5 approximately. This indicates that the acceptance rate is reasonable. However, it is evident that a fixed value for Σ_b does not make sense as it does not converge. Possible improvements to the sampling algorithm, such as exploring new step sizes for the ρ and σ_b^2 random walk algorithms or increasing the number of iterations, could be attempted to have better convergence. However, the current model gave sufficient results to work with it.

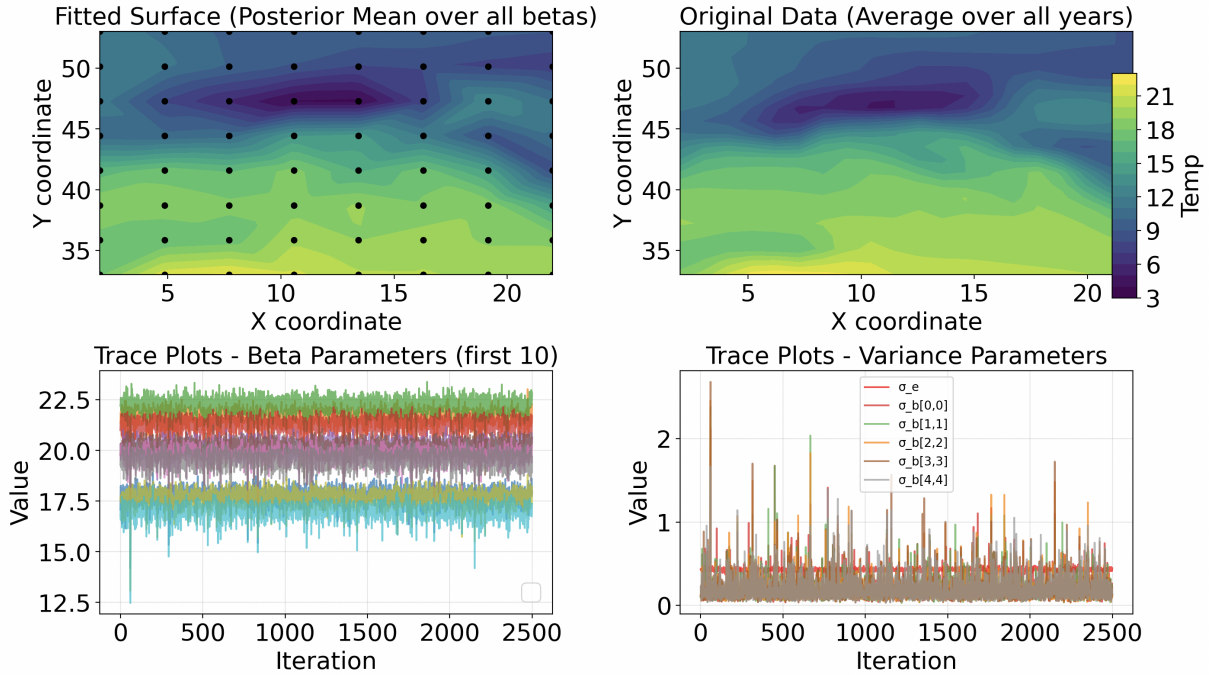


Figure 5: MCMC Results - Inverse Wishart

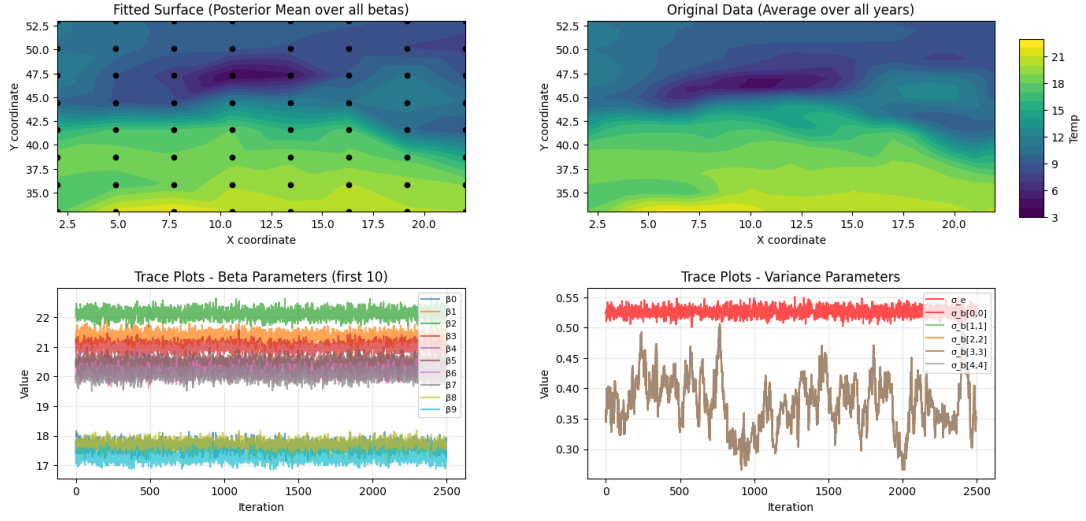


Figure 6: MCMC Results - Metropolis-Hastings

Moreover, the structure and values of the fitted surfaces for individual years (Fig 7 top) strongly match the yearly original data (Fig 7 bottom). This is a great sign of a good-fit model over all years.

Comparing the results in Figures 7 and 8, we observe that the fitted curves for individual years are very similar, with the Matérn covariance matrix having the edge. With proper scaling, the Matérn version is expected to perform more efficiently than Inverse Wishart.

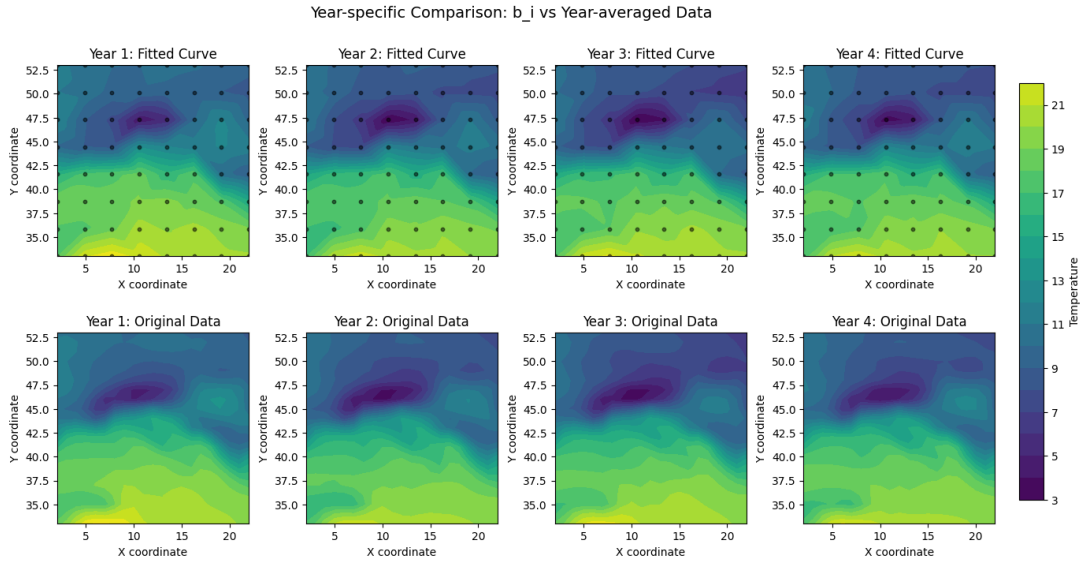


Figure 7: Individual Plots with Inverse Wishart as Covariance Matrix

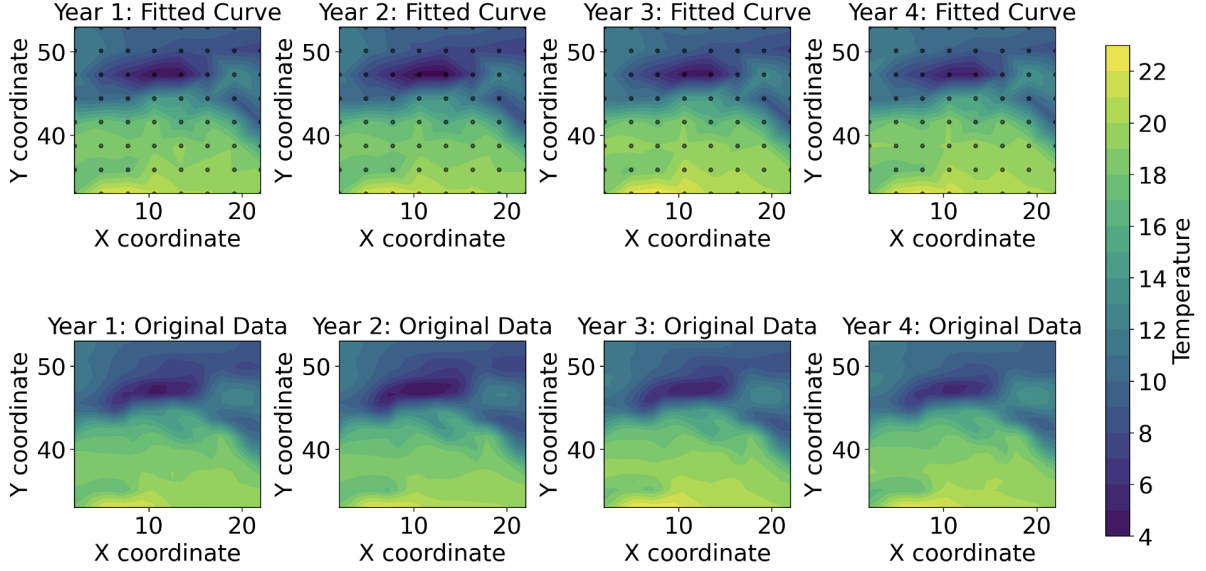


Figure 8: Individual Plots with Matérn as Covariance Matrix

4.7.2 Results covariance matrix

The covariance matrix for the samples when we use the Inverse Wishart distribution to sample σ_b is shown in Figure 9. The most obvious pattern is that the samples have really high covariances between themselves. However, for every other pair of samples, the covariance is near 0, indicating that there is no correlation between the temperature values of two different locations. This makes sense as the Inverse Wishart distribution does not account for the locality in the model.

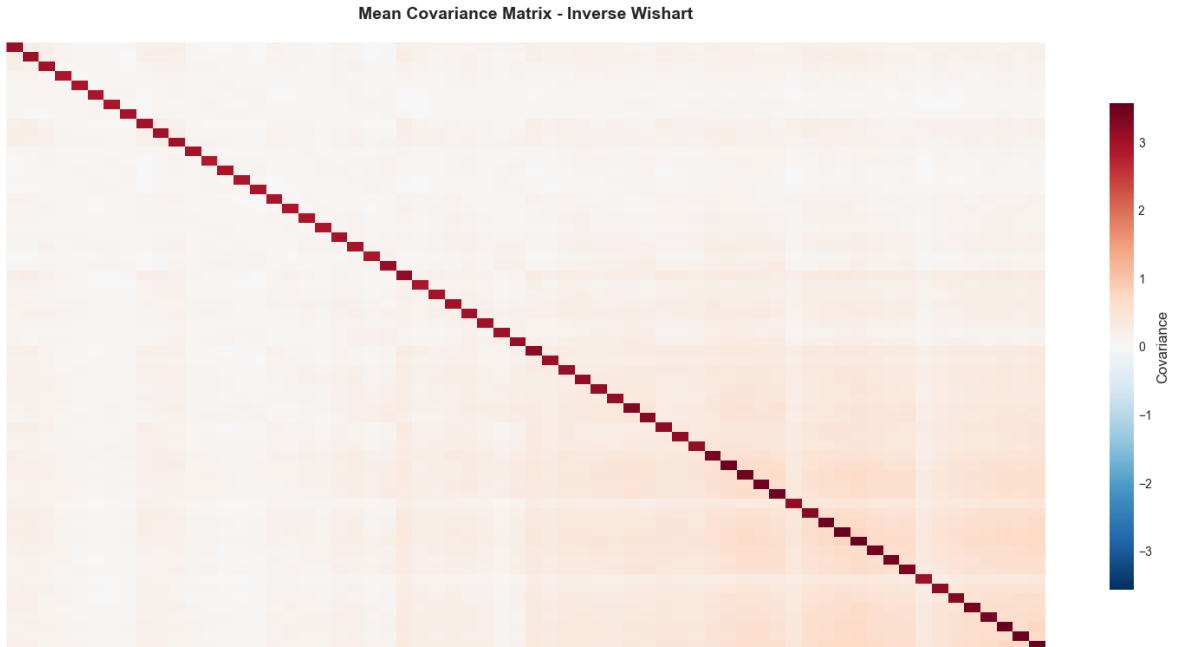


Figure 9: Covariance Matrix for Σ_b in 2D Data - Inverse Wishart

When Matérn covariance matrix with $\nu = 3/2$ is used for the model, the covariance values between nearby locations increase significantly. Two points that neighbour each other have large covariance values, and the correlation gradually decreases as the points get further away, giving

the matrix blurry visuals. The grids in the matrix are the result of how the data is represented in the code. Instead of using a two-dimensional data structure to store longitude latitude values, one vector represented the dataset. Because of this, consecutive latitude values are 20 blocks away from each other in the vector, accounting for the off-diagonal blocks in the matrix.

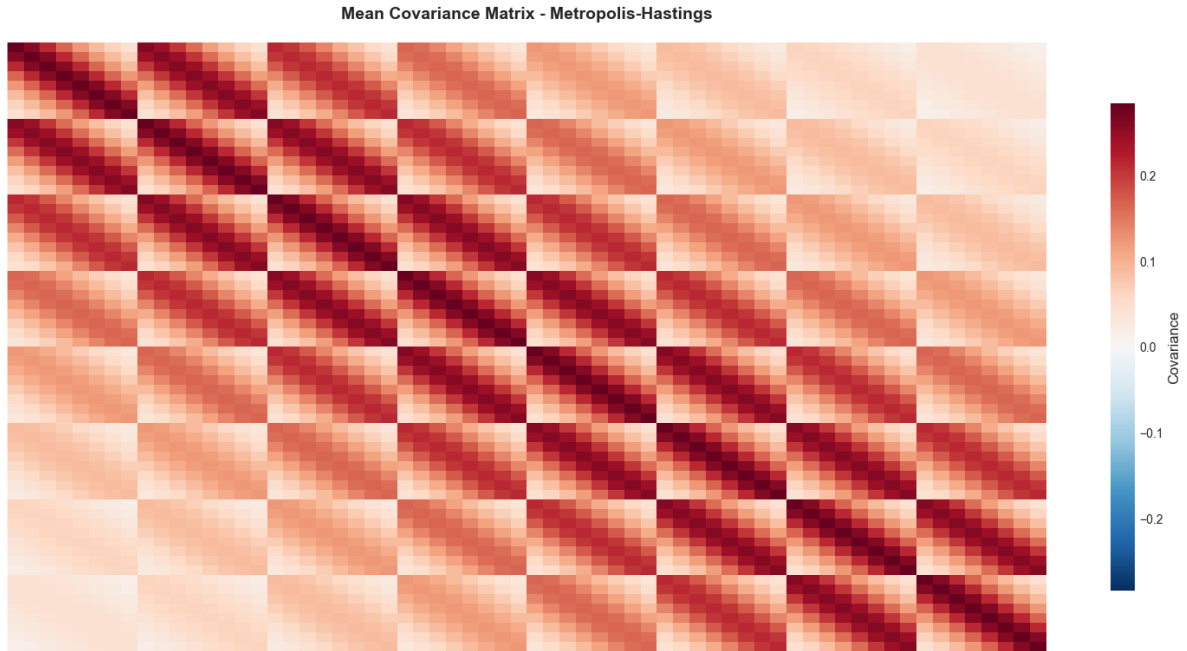


Figure 10: Covariance Matrix for Σ_b in 2D Data - Metropolis-Hastings

5 Conclusion

This study demonstrates that Bayesian hierarchical functional modelling provides an effective probabilistic framework for reconstructing spatial temperature fields from discretely observed climate data. By representing yearly temperature profiles as linear combinations of spatial basis functions with random coefficients, the model successfully separates the global mean spatial trend from year-specific variability, allowing the information from multiple years (1983–2007) to be jointly exploited.

In the one-dimensional setting, cubic B-spline basis functions yield smooth functional approximations of longitudinal temperature profiles along the equator. The posterior mean curves shown in this report closely track the observed data with limited dispersion across posterior samples, indicating that the hierarchical structure effectively reduces noise while preserving large-scale spatial features. Furthermore, the trace plots confirm satisfactory convergence of the Gibbs sampler and relatively low posterior uncertainty in both fixed and random effects parameters.

Extending the framework to two spatial dimensions using a finite element basis enables modelling of spatial temperature variation over latitude–longitude domains.

We do this by defining FEM basis as our 2D basis functions. This causes the number of basis functions to scale as K^2 , and the covariance matrix Σ_b scales quadratically with that again, making direct inversion and determinant computation infeasible ($O(N^3)$ complexity). We adopt a Matérn covariance structure for dealing with this issue. Instead of conditioning on all other points, we condition only on the m -nearest neighbours. This yields a sparse approximation of Σ_b , making it feasible to model the full 2D surface efficiently. However, unlike Inverse Wishart, posterior of σ_b is not semi-conjugate, so we use a Random Walk Metropolis-Hastings (RW-MH) inside the Gibbs Sampler to sample from the distribution of Σ_b .

RW-MH sampling of Σ_b matrices is feasible yet performs poorly in comparison to the Inverse

Wishart distribution. Future work should develop a more efficient sampling algorithm than RW-MH and apply NNGP.

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